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Expt.No: 5

Title: Python program to perform Basic operations on an Image¶

(i) Chaging Contrast and Brightness (ii) Edge Detection, Blur, Sharpening

```
In [4]: 1 import cv2
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 image = cv2.imread('ResizeImage3.jpg')
6
7 adjusted_image = cv2.convertScaleAbs(image, alpha=1.5, beta=50)
8
9 # (ii) Edge Detection
10 edges = cv2.Canny(cv2.cvtColor(image, cv2.COLOR_BGR2GRAY), 100, 200)
11
12 # (iii) Apply Gaussian Blur
13 blurred_image = cv2.GaussianBlur(image, (21, 21), 0)
14
15 # (iv) Sharpen the image using a kernel
16 sharpened_image = cv2.filter2D(
17     image,
18     -1,
19     np.array([[ -1, -1, -1],
20               [-1,  9, -1],
21               [-1, -1, -1]]))
22 )
23
24 plt.figure(figsize=(12, 8))
25
26 plt.subplot(2, 3, 1)
27 plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
28 plt.title('Original Image')
29 plt.axis('off')
30
31 plt.subplot(2, 3, 2)
32 plt.imshow(cv2.cvtColor(adjusted_image, cv2.COLOR_BGR2RGB))
33 plt.title('Adjusted Image (Contrast & Brightness)')
34 plt.axis('off')
35
36 plt.subplot(2, 3, 3)
37 plt.imshow(edges, cmap='gray')
38 plt.title('Edge Detection')
39 plt.axis('off')
40
41 plt.subplot(2, 3, 4)
42 plt.imshow(cv2.cvtColor(blurred_image, cv2.COLOR_BGR2RGB))
43 plt.title('Blurred Image')
44 plt.axis('off')
45
46 plt.subplot(2, 3, 5)
47 plt.imshow(cv2.cvtColor(sharpened_image, cv2.COLOR_BGR2RGB))
48 plt.title('Sharpened Image')
49 plt.axis('off')
50
51 plt.show()
52
53 cv2.imwrite('Adjusted_Image.jpg', adjusted_image)
54 cv2.imwrite('Edge_Detection_Image.jpg', edges)
55 cv2.imwrite('Blurred_Image.jpg', blurred_image)
56 cv2.imwrite('Sharpened_Image.jpg', sharpened_image)
```

Original Image



Adjusted Image (Contrast & Brightness)



Edge Detection



Blurred Image



Sharpened Image



Out[4]: True

In []:

1